SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of ArtificialIntelligence and Machine Learning

ASSESSMENT2 – Scenario-Based Questions (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 2: Scenario-Based Questions (5 Questions × 10 Marks)
Weightage:	35% of CIA	**Total Marks:**	
CO2 Statement:	***Apply image enhancement, filtering, a techniques to analyze images, detect ed boundaries, and solve real-world Comp problems.***		
Student Name:	**S.Venkateswarlu**	**Reg. No.:**	**192425340**
Section / Batch:	**2024**	**Date:**	

Code of Conduct

I S.Venkateswarlu/192425340(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *S.Venkateswarlu*

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Scenario-Based Qns

Q1. Image Enhancement Using Grey-Level Transformations

A hospital uses a digital imaging system to capture X-ray images, but some images appear too dark to identify important anatomical structures. The image processing team decides to apply grey-level transformation techniques to improve image visibility. Apply a suitable transformation to enhance the brightness and contrast of the image. Calculate the transformed intensity values for selected pixels using the chosen transformation. Explain how the transformation improves the visibility of important features in the X-ray image.

Q2. Histogram Processing and Spatial Filtering

A security surveillance system captures images at night, resulting in poor contrast and unwanted noise. The system must enhance the images to identify people and objects more clearly. Apply histogram equalization to improve the contrast of the captured image. Use an appropriate spatial filtering technique to reduce noise while preserving important details. Compare the effects of both techniques and explain how they improve the quality of the surveillance images.

Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

A satellite imaging system captures land images containing noise and blurred boundaries between different geographical regions. The image processing team decides to use frequency domain filtering to improve image quality. Apply a low-pass filter to reduce high-frequency noise and smooth the image. Use a high-pass filter to enhance edges and sharpen important geographical features. Explain the differences between the two filters and determine how each technique contributes to image interpretation.

Q4. Image Segmentation Using Edge Detection and Thresholding

An automated manufacturing system must inspect metal components to identify cracks and surface defects. The captured images contain variations in brightness, making defect detection difficult. Apply an appropriate edge detection technique to identify the boundaries of possible cracks. Use thresholding to separate defective regions from the background and construct a binary image representing the detected defects. Explain how the selected techniques help the inspection system identify defective components.

Q5. Region-Based Segmentation and Boundary Detection

An agricultural monitoring system uses aerial images to identify crop fields, water bodies, and surrounding land regions. The captured images contain regions with similar intensity values and some discontinuous boundaries. Apply region-based segmentation to group neighbouring pixels with similar characteristics. Use edge linking and boundary detection techniques to connect broken edges and identify the boundaries of different regions. Explain how these techniques help distinguish agricultural fields and improve the accuracy of land-region identification.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator Program Director
Signature:	Signature: Signature:
Date:	Date: Date:

Q1. Image Enhancement Using Grey-Level Transformations

Answer:

A hospital may capture X-ray images that are too dark, making bones and other anatomical structures difficult to observe. A suitable method is the linear contrast-stretching transformation. It maps the original intensity range to a wider range so that dark and bright details become easier to distinguish.

The transformation is:

$s = (r - r_{min}) \times (L - 1) / (r_{max} - r_{min})$

Here, r is the original pixel intensity, s is the transformed intensity, r_min and r_max are the minimum and maximum input intensities, and L = 256 for an 8-bit grayscale image. For an illustrative image with r_min = 40 and r_max = 160:

$s = (r - 40) \times 255 / 120$

Calculation for selected pixels:

Original intensity (r)	Calculation	New intensity (s)
40	(40 − 40) × 255 / 120	0
70	(70 − 40) × 255 / 120 = 63.75	64
100	(100 − 40) × 255 / 120 = 127.5	128
130	(130 − 40) × 255 / 120 = 191.25	191
160	(160 − 40) × 255 / 120	255

The example values are illustrative; actual values depend on the original X-ray image. The transformation expands the intensity range from 40–160 to approximately 0–255. This increases contrast between structures that previously had similar gray levels. As a result, bones, tissue boundaries, and other details may become more visible. The enhanced image should still be reviewed by a qualified medical professional.

Q2. Histogram Processing and Spatial Filtering

Answer:

Night-time surveillance images often have low contrast and contain noise caused by poor lighting and camera sensors. Histogram equalization and spatial filtering can be applied in sequence to improve visibility.

1. Histogram equalization

A histogram shows how frequently each gray-level intensity occurs. Histogram equalization uses the cumulative distribution of pixel intensities to redistribute gray levels across a wider range. For an 8-bit image, a common

mapping is:

$s_k = \text{round}[255 \times \text{CDF}(r_k)]$

where CDF(r_k) is the cumulative distribution function for intensity r_k. The process can improve global contrast and make people or objects easier to distinguish from the background. It may, however, also make some noise more visible.

2. Spatial filtering to reduce noise

A 3 × 3 median filter is suitable when the image contains isolated bright or dark noise. For each pixel, the filter takes the nine values in its 3 × 3 neighbourhood, sorts them, and replaces the centre pixel with the median value. For example, for neighbourhood values 20, 21, 19, 22, 200, 20, 18, 21, and 19, the sorted middle value is 20; therefore, the centre value 200 is replaced by 20.

Comparison and result

Histogram equalization improves contrast but does not directly remove noise. Median filtering reduces impulse noise while generally preserving edges better than a simple averaging filter. A practical workflow is to apply mild denoising first and then perform histogram equalization, checking that useful details are not lost. Together, these methods can make people, vehicles, and other objects clearer in surveillance images.

Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

Answer:

Frequency-domain filtering processes an image after converting it into its frequency representation, commonly using the two-dimensional Discrete Fourier Transform (DFT). Low spatial frequencies represent slowly changing regions, while high spatial frequencies represent rapid intensity changes such as edges and fine details, as well as some noise.

1. Low-pass filter

A low-pass filter allows low frequencies to pass and reduces high frequencies. A Gaussian low-pass filter is often useful because it smooths gradually and tends to avoid the strong ringing associated with an abrupt ideal cutoff. The steps are: (i) compute the DFT of the image, (ii) multiply the frequency spectrum by the low-pass filter, and (iii) apply the inverse DFT to obtain the smoothed image.

Effect: random high-frequency noise is reduced and large land regions appear smoother. However, excessive smoothing may blur coastlines, roads, field borders, and other small features.

2. High-pass filter

A high-pass filter suppresses low frequencies and retains or emphasizes high frequencies. It highlights rapid intensity changes, so boundaries between geographical regions and fine linear features become more visible. The filtered result can be used directly for edge emphasis or combined with the original image for sharpening.

Effect: edges and boundaries become clearer, but high-frequency noise may also be amplified. Filtering strength should therefore be selected carefully.

Difference and contribution

Feature	Low-pass filter	High-pass filter
Frequency action	Keeps low frequencies; reduces high frequencies	Reduces low frequencies; keeps/emphasizes high frequencies
Main purpose	Smoothing and noise reduction	Edge enhancement and sharpening
Possible limitation	Can blur boundaries	Can amplify noise

Using both methods carefully helps analysts view broad land patterns and inspect boundaries. The low-pass result supports interpretation of larger regions, while the high-pass result helps locate edges and small geographical features.

Q4. Image Segmentation Using Edge Detection and Thresholding

Answer:

In metal-component inspection, cracks may appear as narrow lines or intensity changes. Edge detection can locate possible crack boundaries, while thresholding can separate candidate defects from the background.

1. Edge detection

The Canny edge detector is a suitable choice because it smooths the image, calculates intensity gradients, applies non-maximum suppression, and uses two thresholds with edge tracking by hysteresis. It can produce relatively thin and connected edges. A small Gaussian smoothing step can reduce noise before edge detection.

2. Thresholding and binary segmentation

After edge detection, a threshold can be applied to a grayscale or edge-strength image. For a grayscale defect image, a simple binary threshold is:

$B(x, y) = 1$ if $I(x, y) < T$; otherwise $B(x, y) = 0$

This example assumes cracks are darker than the background. If defects are brighter, the comparison is reversed.

T is the selected threshold, I(x, y) is the pixel intensity, and B(x, y) is the binary output. For example, if T = 80, pixels with intensity 35 or 70 become 1 (candidate defect), while pixels with intensity 100 or 150 become 0 (background). The exact threshold must be chosen from the image; adaptive thresholding may be more appropriate when illumination varies across the component.

3. Constructing and interpreting the binary image

The binary image marks candidate defective pixels as white (1) and background pixels as black (0), or vice versa depending on the display convention. Small isolated noise regions can be removed using morphological opening, and small gaps can sometimes be connected using morphological closing. Connected-component analysis can then measure each candidate region.

The system can flag a component when detected regions have crack-like length, width, or shape. Edge detection provides boundary information, and thresholding separates candidate regions from the background. Combining them improves automated inspection, but the final decision should use suitable thresholds and validation to avoid false alarms or missed defects.

Q5. Region-Based Segmentation and Boundary Detection

Answer:

Aerial agricultural images contain crop fields, water bodies, and other land regions. Region-based segmentation groups neighbouring pixels that have similar properties, while edge linking and boundary detection help identify the outlines of these regions.

1. Region-based segmentation

Region growing is one suitable technique. First, select seed pixels inside representative regions such as a crop field or water body. Then examine neighbouring pixels and add a pixel to a region when its intensity, colour, or texture is sufficiently similar to the region's properties. Continue until no eligible neighbouring pixels remain. Separate seeds can be used for different land-cover types.

The similarity rule may use a threshold such as the absolute difference between a neighbouring pixel's intensity and the region mean. A strict threshold may split one field into several regions, while a loose threshold may merge different land types. Seeds and thresholds should therefore be selected carefully.

2. Edge linking and boundary detection

An edge detector can identify local changes in intensity, but shadows, vegetation, and noise may produce broken edges. Edge linking connects nearby edge pixels when their positions, orientations, and gradient directions are

consistent. Morphological operations or contour-tracing methods can also help close small gaps. Boundary detection then traces the connected outlines of the segmented regions.

3. How the techniques work together

Region-based segmentation groups pixels into meaningful areas, while edge linking connects discontinuous outlines and boundary detection identifies the final region borders. For example, a water body may form a region with relatively consistent colour, while its boundary separates it from surrounding crops. Combining colour or intensity similarity with connected boundaries helps distinguish fields, water bodies, and other land regions. Accuracy can be improved by using suitable seeds, thresholds, and image preprocessing, followed by checking the segmentation against known or manually labelled regions.

Conclusion

Image enhancement improves visibility, histogram equalization improves contrast, spatial filtering reduces noise, frequency-domain filters support smoothing and sharpening, and segmentation separates meaningful regions from backgrounds. The appropriate technique depends on the image characteristics and the goal of the computer vision application.